

## GradeSense — Annotated Answer

**Emergency Room Wait Time Prediction: An Analytical and Predictive Modeling Approach for Healthcare Optimization** Namami Verma, Payel Manna, Ardhi Jasmine, Kavya Raghavendran, and Anika Tripathi Team Data Walkers October 4, 2025 Abstract The optimization of patient flow within hospital Emergency Rooms (ERs) is a critical challenge in modern healthcare, directly impacting patient outcomes, satisfaction, and operational efficiency. This research presents the development and evaluation of a machine learning framework designed to analyze and predict ER wait times using historical patient data. A dual-methodology approach was employed, addressing the problem from both a regression perspective—predicting the exact wait time in minutes—and a classification perspective—prioritizing patients by categorizing them into low-wait or high-wait cohorts. Various models were evaluated, with the Random Forest algorithm demonstrating superior performance for both tasks. The final regression model achieved a coefficient of determination ( $R^2$ ) of 0.77 and a Mean Absolute Error (MAE) of 24.54 minutes, indicating a strong predictive capability. The classification model reliably identified patients at risk of long waits, enabling proactive resource allocation. Feature importance analysis revealed that the time elapsed until a patient is seen by a medical professional is the single most significant predictor of total wait time. These findings provide actionable, data-driven insights for hospital administrators to streamline ER processes, reduce bottlenecks, and enhance the quality of patient care.

-- 1 of 6 --

### 1 Introduction

#### 1.1 Background and Motivation

Emergency Rooms are high-pressure environments that serve as the primary entry point for a significant volume of patients requiring urgent medical care. Overcrowding and prolonged wait times are persistent issues that can lead to adverse patient outcomes, decreased patient satisfaction, and increased operational strain on healthcare providers. The ability to accurately predict patient wait times is not merely a logistical advantage; it is a crucial component of modern, patient-centric healthcare delivery. Predictive analytics offers a powerful tool to transform ER management from a reactive to a proactive model, allowing for dynamic resource allocation, improved patient communication, and targeted process improvements.

#### 1.2 Problem Statement

The primary objective of this research is to develop a robust predictive model that can accurately estimate ER wait times. To achieve a comprehensive solution, this problem is bifurcated into two distinct, yet complementary, analytical tasks:

- \* **The Regression Problem:** To predict the exact total wait time (in minutes) for a patient from arrival to departure. This continuous variable prediction is based on the timestamps recorded at various stages of the ER journey, including registration, triage, diagnostics, and consultation.
- \* **The Classification Problem:** To categorize patients into discrete wait-time groups (e.g., 'low-wait' or 'high-wait') based on a predetermined threshold, such as the median wait time. This approach aims to identify at-risk patients early, facilitating triage and resource prioritization.

#### 1.3 Research Objectives

- \* To preprocess and engineer features from a real-world ER dataset to prepare it for machine learning.
- \* To evaluate a suite of machine learning algorithms for both regression and classification tasks.
- \* To identify the best-performing model based on rigorous statistical metrics such as  $R^2$ , Mean Absolute Error (MAE), and classification accuracy.
- \* To conduct a feature importance analysis to identify the primary drivers and bottlenecks within the ER workflow.
- \* To translate model outputs into actionable insights for hospital administrators.

1 -- 2 of 6 -- 2

**Methodology**

### 2.1 Data Source and Description

The study utilized a publicly available dataset from Kaggle, containing anonymized historical records of ER visits. The dataset encompasses a rich set of features, including unique identifiers for visits, patients, and hospitals; timestamps for key process stages (e.g., arrival, triage, consultation); patient urgency levels; and visit dates.

### 2.2 Data Preprocessing and Feature Engineering

A multi-stage data preparation pipeline was constructed to ensure data quality and model compatibility:

- Data Cleaning:** Initial exploratory data analysis was performed to identify and handle missing values and inconsistencies. Non-predictive columns, such as Visit ID, Hospital Name, and Patient ID, were removed to reduce model noise and protect patient privacy.
- Temporal Feature Extraction:** The Visit Date column was converted into a standardized datetime format. From this, several temporal features were engineered to capture cyclical patterns in ER traffic, including the hour of the day, day of the week, month, and a binary weekend indicator.
- Categorical Variable Encoding:** Categorical features, such as the urgency level, were transformed into a numerical format using one-hot encoding.
- Data Splitting and Scaling:** The dataset was partitioned into a training set (80%) and a testing set (20%). For distance-based algorithms like K-Nearest Neighbors (KNN), features were normalized using a standard scaler.

### 2.3 Modeling and Evaluation

#### 2.3.1 Regression Modeling

- Models Evaluated:** Linear Regression, Ridge Regression, Lasso Regression, K-Nearest Neighbors (KNN), Decision Tree, and Random Forest.
- Performance Metrics:**
  - Mean Absolute Error (MAE):** The average absolute difference between predicted and actual values.  $MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$
  - Root Mean Squared Error (RMSE):** The square root of the average squared differences, penalizing larger errors more heavily.  $RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$
  - Coefficient of Determination (R<sup>2</sup>):** The proportion of variance in the dependent variable predictable from the independent variables.

#### 2.3.2 Classification Modeling

- Models Evaluated:** K-Nearest Neighbors (KNN), Decision Tree, and Random Forest.
- Performance Metrics:** Performance was assessed using confusion matrices, overall accuracy, precision, and recall.

## 3 Results and Analysis

### 3.1 Regression Performance

The Random Forest Regressor emerged as the top-performing model. Table 1 summarizes its performance on the test dataset.

| Performance Metric | Value                    | Interpretation                                                                                                                       |
|--------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| MAE                | 24.54 min                | The model's predictions are, on average, within approximately 25 minutes of the actual wait time. This is operationally significant. |
| MSE                | 1526.10 min <sup>2</sup> | The Mean Squared Error, which heavily penalizes large errors.                                                                        |
| RMSE               | 39.06 min                | The square root of MSE, indicating the standard deviation of the prediction errors.                                                  |
| R <sup>2</sup>     | 0.77                     | The model successfully explains 77% of the variability in ER wait times.                                                             |

### 3.2 Classification Performance

The Random Forest Classifier also demonstrated superior performance. It achieved a high true positive rate for the 'high-wait' class while maintaining a low false positive rate, making it a reliable tool for proactive operational management.

### 3.3 Feature Importance Analysis

An analysis of the feature importances from the Random Forest model provided clear insights:

- Time to Medical Professional (40-50% Importance):** Overwhelmingly the most influential factor.
- Time to Triage (20-30% Importance):** The second most significant contributor.
- Temporal Features (10-20% Importance):** Collectively held moderate importance. This analysis indicates that interventions aimed at reducing the time to see a medical professional will yield the most significant impact.

## 4 Discussion

### 4.1 Interpretation of Findings

The R<sup>2</sup> value of 0.77 confirms that a significant portion of ER wait time variance is systematic and predictable. The MAE of 24.54 minutes is well within an operationally

useful range. The feature importance results empirically validate that the primary bottleneck is access to core clinical resources.

#### 4.2 Strengths and Limitations

- \* Strengths: High predictive accuracy, actionable insights, and robustness of the ensemble method.
- \* Limitations: The model is analytical rather than real-time predictive at arrival; performance is constrained by available features (e.g., no data on staffing levels); generalizability requires recalibration for different hospitals.

### 5 Conclusion and Future Work

#### 5.1 Conclusion

This research successfully developed a robust machine learning solution for analyzing and predicting ER wait times. By achieving high accuracy ( $R^2 = 0.77$ , MAE=24.54 minutes) and providing interpretable insights into process bottlenecks, the model serves as a powerful decision-support tool. It provides a data-driven foundation for optimizing patient flow and unequivocally identifies the time to consultation as the most critical area for improvement.

#### 5.2 Recommendations for Future Work

- \* Model Enhancement: Explore gradient boosting machines (e.g., XGBoost) and conduct systematic hyperparameter optimization.
- \* Feature Engineering: Develop an arrival-time prediction model using only features available at registration. Incorporate external data sources like weather and public health surveillance.
- \* Deployment: Create a deployment-ready API to integrate predictions into existing Hospital Information Systems (HIS) and evolve the model towards prescriptive analytics.
- \* Ethical Considerations: Conduct a fairness and bias audit to ensure the model performs equitably across different patient demographics.

## Corrections & Feedback

1. [General] Edited note - testing independence from grading
2. [General] [Explains how/why it works] No evidence found for this rubric point.
3. [General] [Gives a correct, relevant example] No evidence found for this rubric point.
4. [General] [Clarity and structure of the answer] No evidence found for this rubric point.